Solution of Triangle: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Triangle basics

KEY POINTS

- $A + B + C = \pi$; side a is opposite angle A
- Triangle inequality: $|b - c| < a < b + c$ for every side
- Larger side $\Leftrightarrow$ larger opposite angle
- $2s = a + b + c$; Δ = area; R, r, r_1, r_2, r_3 = circum-, in-, ex-radii
- Solving a triangle needs three independent elements, at least one a side

2. Sine rule

Formulas

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

KEY POINTS

- $a = 2R \sin A$: converts sides to angles in identities
- Angles known $\Rightarrow a : b : c = \sin A : \sin B : \sin C$
- Use with one side and two angles; in two sub-triangles joined by a cevian
- $\sin A \sin(B - C) = \sin^2 B - \sin^2 C$

3. Cosine rule and nature

Formulas

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \qquad \cot A = \frac{b^2 + c^2 - a^2}{4\Delta}$$

KEY POINTS

- Use when three sides, or two sides and the included angle, are known
- Sides in ratio: take $a = pk$ etc.; k cancels
- c largest: $a^2 + b^2 \gtrless c^2$ gives acute / right / obtuse
- $\sum \cot A = \dfrac{a^2 + b^2 + c^2}{4\Delta}$

Watch out

$\cos C < 0$ only for the largest angle; check the largest angle first

4. Projection and tangent rules

Formulas

$$a = b \cos C + c \cos B$$

$$\tan \frac{B - C}{2} = \frac{b - c}{b + c} \cot \frac{A}{2}$$

KEY POINTS

- Projection: for sums like $b \cos C + c \cos B$; write $\cos^2 \frac{C}{2} = \frac{1 + \cos C}{2}$ first
- Tangent rule: two sides + included angle, asked $B - C$ or the other angles

5. Area of triangle

Formulas

$$\Delta = \tfrac{1}{2} bc \sin A = \frac{abc}{4R} = rs = \sqrt{s(s - a)(s - b)(s - c)}$$

KEY POINTS

- Altitude on a: $p_a = \dfrac{2\Delta}{a}$; largest altitude on smallest side
- Triangles with a common angle: area ratio = product of ratios of the enclosing sides
- Split Δ into parts from an inner point ($r = \frac{\Delta}{s}$, circle with centre on a side)

6. Half-angle formulae

Formulas

$$\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}},\ \cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}},\ \tan\frac{A}{2} = \frac{\Delta}{s(s-a)}$$

KEY POINTS

- $\tan\frac{A}{2}\tan\frac{B}{2} = \frac{s-c}{s}$; a, b, c in A.P. $\Leftrightarrow \tan\frac{A}{2}\tan\frac{C}{2} = \frac{1}{3}$
- $(s-a) : (s-b) : (s-c)$ given $\Rightarrow$ use these forms directly
- $\sum \cot\frac{A}{2} = \frac{s^2}{\Delta}$

7. Ambiguous case (b, c, B given)

Formulas

$$a = c\cos B \pm \sqrt{b^2 - c^2\sin^2 B}$$

STEPS

1. $b < c\sin B$: no triangle
2. $b = c\sin B$: one (right-angled)
3. $c\sin B < b < c$: two triangles
4. $b \geq c$: one triangle

Watch out

Keep only positive roots for a

8. m-n theorem

Formulas

$$(m+n)\cot\theta = m\cot\alpha - n\cot\beta = n\cot B - m\cot C$$

KEY POINTS

- $BD : DC = m : n$, $\theta = \angle ADC$, $\alpha = \angle BAD$, $\beta = \angle DAC$
- Median: $2\cot\theta = \cot\alpha - \cot\beta = \cot B - \cot C$

Watch out

θ is the angle on the side of C ($\angle ADC$), not $\angle ADB$

9. Circumcircle

Formulas

$$R = \frac{a}{2\sin A} = \frac{abc}{4\Delta}$$

KEY POINTS

- O = meet of perpendicular bisectors; $OA = OB = OC = R$
- Distance of O from BC: $R\cos A$; $\angle BOC = 2A$
- Inside (acute), midpoint of hypotenuse (right), outside (obtuse)

10. Incircle

Formulas

$$r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2} = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

KEY POINTS

- I = meet of internal bisectors; always inside; distance r from each side
- $IA = r\operatorname{cosec}\frac{A}{2}$; $\angle BIC = \frac{\pi}{2} + \frac{A}{2}$
- Tangent lengths: $s-a$, $s-b$, $s-c$ from A, B, C
- Bisector: $BD : DC = c : b$; $AI : ID = (b+c) : a$
- Right angle at C: $r = s - c$

11. Excircles

Formulas

$$r_1 = \frac{\Delta}{s-a} = s\tan\frac{A}{2} = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$$

KEY POINTS

- I_1 = internal bisector of A + external bisectors of B, C; outside, beyond BC
- $I_1A = r_1\operatorname{cosec}\frac{A}{2}$, $I_1B = r_1\sec\frac{B}{2}$; $\angle BI_1C = \frac{\pi}{2} - \frac{A}{2}$
- $\sum r_1 = 4R + r$, $\sum r_1r_2 = s^2$, $rr_1r_2r_3 = \Delta^2$, $\sum\frac{1}{r_1} = \frac{1}{r}$
- Tangent from A to the excircle opposite A: s

Watch out

r_1 is opposite A: largest side gives largest exradius

12. Median, centroid, angle bisector

Formulas

$$m_a = \tfrac{1}{2}\sqrt{2b^2+2c^2-a^2}, \qquad AD_{\text{bis}} = \frac{2bc}{b+c}\cos\frac{A}{2}$$

KEY POINTS

- G divides each median $2:1$ from the vertex; always inside
- A median halves the area; three medians make six equal areas
- Foot of altitude H' and midpoint M of AB: $H'M = \dfrac{|a^2-b^2|}{2c}$

13. Orthocentre, pedal and excentral triangles

Formulas

$$HA = 2R\cos A, \qquad HD = 2R\cos B\cos C, \qquad \angle BHC = \pi - A$$

KEY POINTS

- Pedal DEF: sides $a\cos A$, $b\cos B$, $c\cos C$; angles $\pi - 2A$ etc.; circumradius $\frac{R}{2}$
- H = incentre of the pedal triangle; pedal area $= 2\Delta\cos A\cos B\cos C$
- Excentral $I_1I_2I_3$: angles $\frac{\pi}{2} - \frac{A}{2}$; sides $4R\cos\frac{A}{2}$; circumradius $2R$

Watch out

Pedal formulas as written are for an acute triangle

14. Regular polygon

Formulas

$$a = 2R\sin\frac{\pi}{n} = 2r\tan\frac{\pi}{n}, \qquad \text{Area} = \frac{n}{2}R^2\sin\frac{2\pi}{n} = nr^2\tan\frac{\pi}{n}$$

KEY POINTS

- Interior angle $\frac{(n-2)\pi}{n}$; diagonal k apart $= 2R\sin\frac{k\pi}{n}$
- $R = r\sec\frac{\pi}{n}$; $R + r = \frac{a}{2}\cot\frac{\pi}{2n}$

15. Equilateral triangle and cyclic quadrilateral

Formulas

$$\Delta = \frac{\sqrt{3}}{4}a^2,\ R = \frac{a}{\sqrt{3}},\ r = \frac{a}{2\sqrt{3}},\ r_1 = \frac{\sqrt{3}}{2}a$$

$$\text{Area}_{\text{cyclic}} = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

KEY POINTS

- Equilateral: $r : R : r_1 = 1 : 2 : 3$
- Cyclic quadrilateral: opposite angles supplementary; split by a diagonal and use the cosine rule twice

16. Common mistakes checklist

STEPS

1. Check the triangle inequality for every answer with sides
2. Two sides + non-included angle: test the ambiguous case
3. $\sin A = \frac{1}{2}$ gives $30°$ or $150°$: use the side order to choose
4. $R\cos A$, $2R\cos A$ become negative for an obtuse A
5. r_1 uses $s - a$, not s; tangent lengths $s - a$ to the incircle, s to the excircle
6. Bisector ratio is $c : b$ (adjacent sides), not $b : c$